



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI CHANDRA CHOODESHWARA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

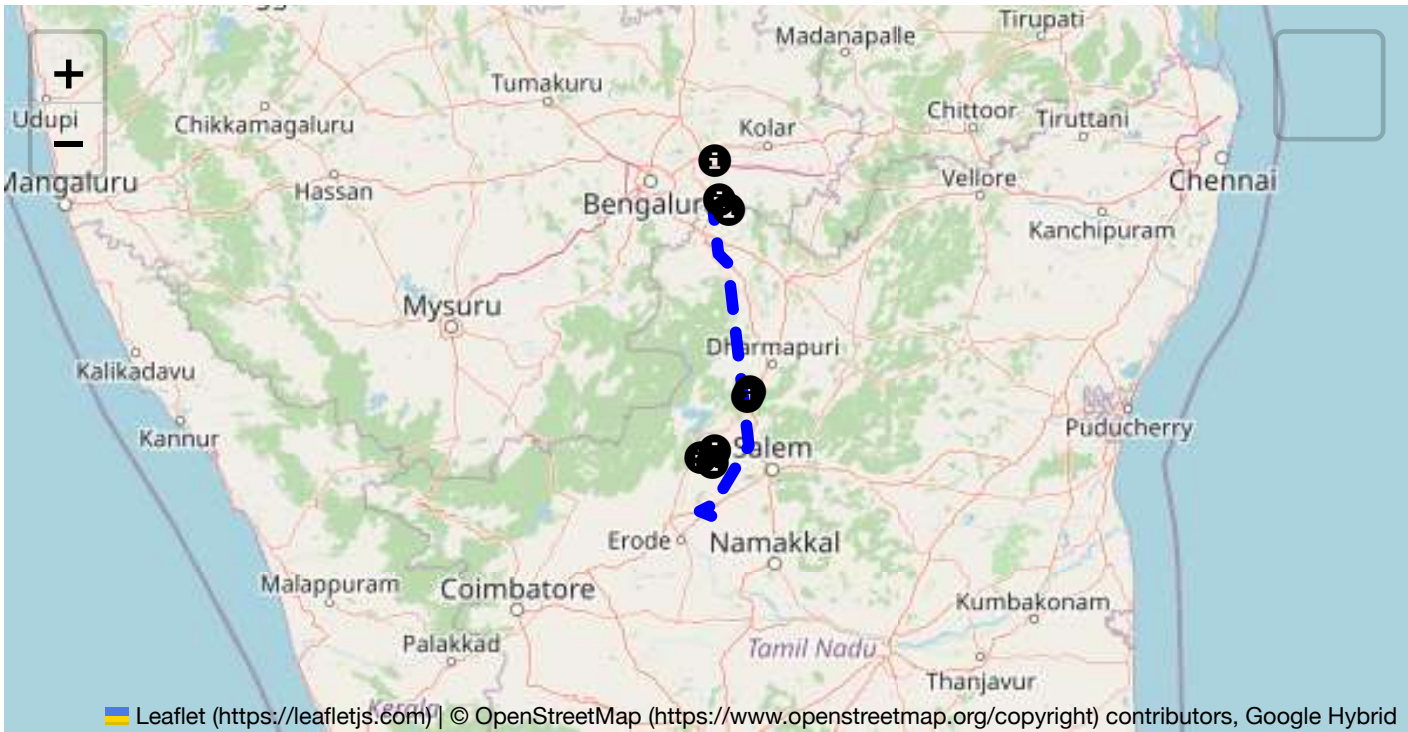
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

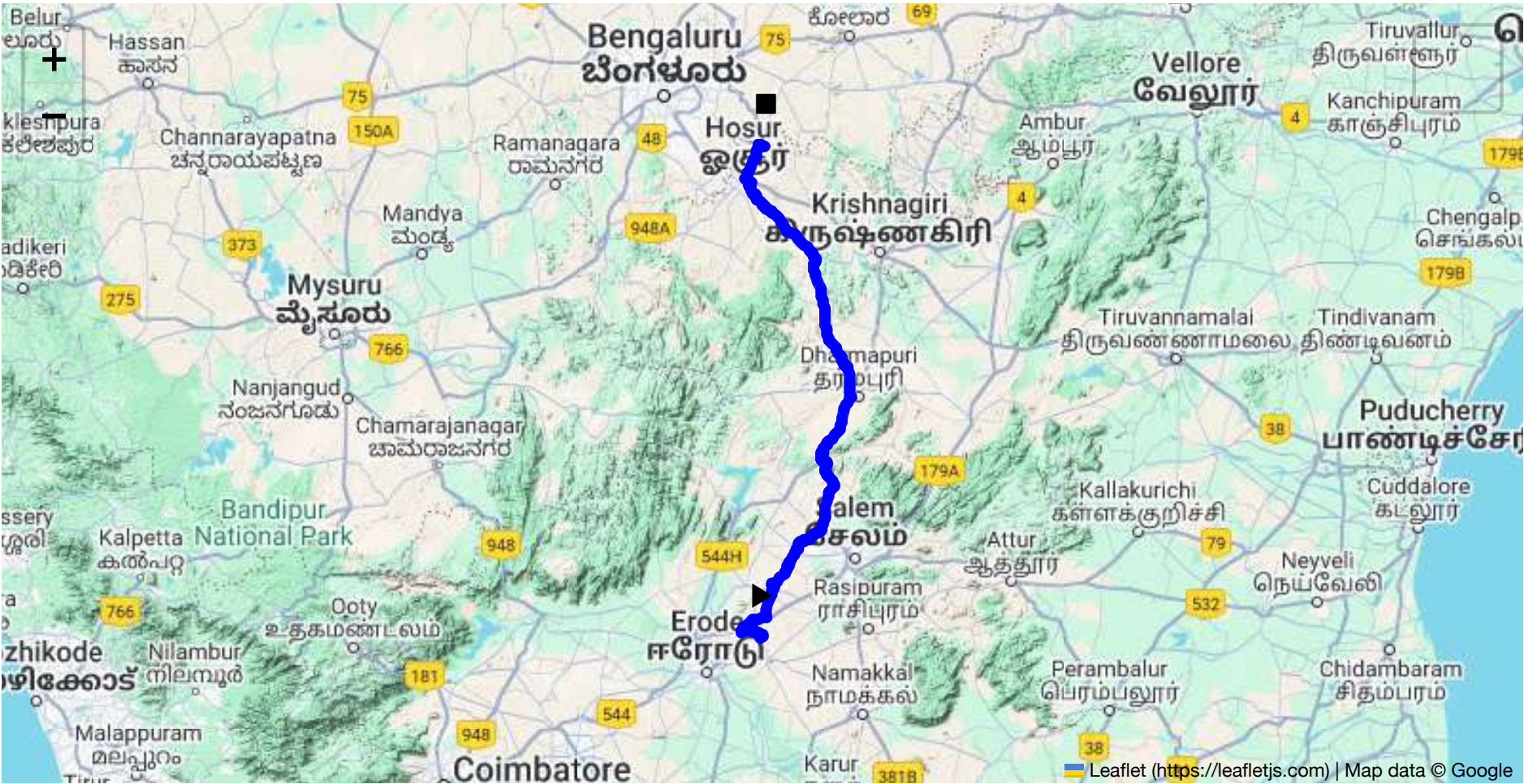


Route Summary:
Total Distance: 199.30 km
Estimated Duration: 4.2 hours

Adjusted Duration (Heavy Vehicle): 5.2 hours

Start: (11.4381, 77.8734)

End: (12.827847, 77.883499)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to RVHM+5C2, Bagalur - Berikai Rd spans approximately 199.30 kilometers and passes through significant areas like Padaiveedu, Ayveli, Salem, Omalur, Kothur, and Thuppuganapalli. The primary roads involved include Bhavani Main Road, SH 86, and SH 17, which are crucial for heavy traffic movement.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- General Climate:** The Tamil Nadu region typically experiences a tropical climate with hot summers, frequent rainfall during the monsoon season, and mild winters.
- Potential Hazards:** Monsoon season (June to September) can lead to heavy rainfall, causing slippery roads, waterlogging, and reduced visibility. Temperature extremes in summer can also lead to heat-related wear on vehicles and tires.

3. Analysis of Traffic Patterns

- Peak Hours:** Morning (7:00 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) are peak traffic hours, especially around Salem and Omalur.
- Congestion-Prone Areas:** Salem and the areas around major intersections can experience significant traffic congestion, particularly during peak hours.

4. Assessment of Road Quality and Infrastructure

- **General Road Conditions:** Bhavani Main Road and SH 86 are mostly well-maintained, though certain rural stretches may have deteriorated sections.
- **Infrastructure Challenges:** Traffic bottlenecks might occur at major junctions and areas where road repairs are underway.

5. Suggestions for Alternative Routes for Emergencies

- **Alternate Route Bypass:** Consider bypassing Salem city traffic by taking nearby secondary roads that are less congested but still adequate for trucks, such as those that link small village detours around major towns.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Transport Regulations:** State and national regulations mandate specific permits and adherence to safety standards for transporting hazardous materials, including secure packaging and signaling.
- **Restricted Areas:** Certain urban zones, especially dense areas in Salem, may have restrictions on transporting hazardous goods during peak hours.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Incidents Record:** There have been minor incidences of vehicle breakdowns and traffic congestion due to heavy vehicles within Salem city limits. Reports of spills or accidents involving hazardous materials are scarce but adhering to safety guidelines reduces risks.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** The route crosses several agricultural zones where awareness and careful transportation are necessary to prevent spillage contamination.
- **Consideration:** Minimize noise and emissions, especially in residential and environmentally sensitive areas.

9. Analysis of Communication Coverage

- **Coverage:** Mobile network coverage is generally good along major highways and within urban areas, but remote stretches between towns may experience spotty service.
- **Dead Zones:** Be cautious of potential dead zones in rural areas like around Kothur.

10. Estimated Emergency Response Times

- **Urban Sections (e.g., Salem):** Emergency services can typically respond within 15-30 minutes.
- **Rural and Remote Areas:** Response times may extend to 45-60 minutes due to limited resources.

11. Overall Summary of Risk Assessment

- **Main Risks:** Heavy traffic and congestion, especially during peak hours; weather-related hazards during the monsoon; occasional road quality issues in rural areas.
- **Risk Mitigation:**

- Plan travel to avoid peak hour congestion.
- Maintain safe driving practices during adverse weather.
- Ensure vehicles are in optimal condition to handle varying road surfaces.
- **Final Notes:** Proper route planning, awareness of road conditions, adherence to transport regulations, and usage of GPS for real-time updates and navigation adjustments are pivotal in minimizing transportation risks on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.83439, 77.86802	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	Medium	11.43995, 77.87431	30 KM/Hr
3	Turn	High	11.44107, 77.87528	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	Medium	11.45353, 77.85697	30 KM/Hr
6	Turn	High	11.46307, 77.84941	15 KM/Hr
7	Turn	Medium	11.45546, 77.81800	30 KM/Hr
8	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
9	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
10	Turn	High	11.49469, 77.88488	15 KM/Hr
11	Turn	High	11.55618, 77.89789	15 KM/Hr
12	Turn	Medium	11.55699, 77.89986	30 KM/Hr
13	Turn	Medium	11.56455, 77.90586	30 KM/Hr
14	Turn	High	11.58228, 77.90841	15 KM/Hr
15	Turn	Medium	11.58308, 77.90714	30 KM/Hr
16	Turn	Medium	11.58563, 77.90571	30 KM/Hr
17	Turn	Medium	11.59201, 77.90649	30 KM/Hr
18	Turn	Medium	11.63882, 77.94859	30 KM/Hr
19	Turn	High	11.64226, 77.95200	15 KM/Hr
20	Turn	Medium	11.64716, 77.95118	30 KM/Hr
21	Turn	Medium	11.67868, 77.96836	30 KM/Hr
22	Turn	Medium	11.69893, 77.97976	30 KM/Hr
23	Turn	High	11.70223, 77.97860	15 KM/Hr
24	Turn	Medium	11.71402, 77.99586	30 KM/Hr
25	Turn	Medium	11.71704, 77.99948	30 KM/Hr
26	Turn	Medium	11.71918, 78.01127	30 KM/Hr
27	Turn	Medium	11.72708, 78.02197	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
28	Turn	High	11.73769, 78.03451	15 KM/Hr
29	Turn	High	11.73651, 78.03871	15 KM/Hr
30	Turn	Medium	11.74143, 78.04613	30 KM/Hr
31	Turn	High	11.86821, 78.08359	15 KM/Hr
32	Turn	Medium	11.87928, 78.08201	30 KM/Hr
33	Turn	Medium	11.91391, 78.05879	30 KM/Hr
34	Turn	Medium	11.92575, 78.06505	30 KM/Hr
35	Turn	Medium	11.93105, 78.06491	30 KM/Hr
36	Turn	Medium	11.93438, 78.06315	30 KM/Hr
37	Turn	Medium	11.93701, 78.05407	30 KM/Hr
38	Turn	Medium	11.95656, 78.04774	30 KM/Hr
39	Turn	High	11.95816, 78.04976	15 KM/Hr
40	Turn	High	11.96390, 78.05138	15 KM/Hr
41	Turn	Medium	11.96495, 78.04990	30 KM/Hr
42	Turn	High	11.96667, 78.04934	15 KM/Hr
43	Turn	Medium	12.11974, 78.13273	30 KM/Hr
44	Turn	Medium	12.50486, 78.03132	30 KM/Hr
45	Blind Spot	Blind Spot	12.51381, 78.03208	10 KM/Hr
46	Turn	Medium	12.51373, 78.03023	30 KM/Hr
47	Turn	Medium	12.51632, 78.02571	30 KM/Hr
48	Turn	Medium	12.52916, 78.02035	30 KM/Hr
49	Turn	Medium	12.65349, 77.88773	30 KM/Hr
50	Turn	Medium	12.65598, 77.88604	30 KM/Hr
51	Turn	Medium	12.65770, 77.88301	30 KM/Hr
52	Turn	Medium	12.69074, 77.85475	30 KM/Hr
53	Turn	Medium	12.69365, 77.85440	30 KM/Hr
54	Turn	Medium	12.71780, 77.83894	30 KM/Hr
55	Turn	Medium	12.71956, 77.83555	30 KM/Hr
56	Turn	High	12.71961, 77.83551	15 KM/Hr
57	Blind Spot	Blind Spot	12.72777, 77.84311	10 KM/Hr
58	Turn	High	12.73486, 77.82986	15 KM/Hr
59	Turn	Medium	12.73651, 77.82982	30 KM/Hr
60	Turn	Medium	12.73956, 77.83270	30 KM/Hr
61	Turn	Medium	12.80956, 77.85773	30 KM/Hr
62	Turn	Medium	12.83351, 77.86931	30 KM/Hr
63	Turn	Medium	12.83283, 77.87378	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
2	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
12	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
15	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
16	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
17	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
18	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
19	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
23	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
24	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
25	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium
28	pharmacy	H2 Pharmacy	12.7301802, 77.8401037	30 km/h	Medium
29	clinic	Meera Nursing Home, Krishnagiri	12.7326185, 77.8292795	30 km/h	Medium
32	hospital	Amma Hospital	12.7393554, 77.8326981	30 km/h	Medium
33	pharmacy	துளசி பார்மசி	12.7457796, 77.8339248	30 km/h	Medium
35	clinic	Thirumala Private Nursing Home	12.835671, 77.8672146	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
21	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium
27	school	Panchayat Union Primary School	12.6078768, 77.9381865	30 km/h	Medium
30	school	Vijay Vidyalaya School	12.7397242, 77.8308427	30 km/h	Medium
31	school	Maharishi Vidhya Mandir Primary School	12.7397254, 77.8324166	30 km/h	Medium
34	school	யோகி வேமனா உயர்நிலை பள்ளி	12.7952478, 77.8563035	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.83439, 77.86802



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44107, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55618, 77.89789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58563, 77.90571



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59201, 77.90649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63882, 77.94859



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64226, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64716, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67868, 77.96836



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69893, 77.97976



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70223, 77.97860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71402, 77.99586



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71704, 77.99948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71918, 78.01127



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72708, 78.02197



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73769, 78.03451



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74143, 78.04613



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.86821, 78.08359



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87928, 78.08201



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91391, 78.05879



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92575, 78.06505



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93105, 78.06491



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93438, 78.06315



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93701, 78.05407



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95656, 78.04774



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95816, 78.04976



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96390, 78.05138



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96495, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96667, 78.04934



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11974, 78.13273



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50486, 78.03132



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.51381, 78.03208



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51373, 78.03023



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51632, 78.02571



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52916, 78.02035



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.65770, 77.88301



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.69074, 77.85475



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.69365, 77.85440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.71780, 77.83894



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.71956, 77.83555



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.71961, 77.83551



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.72777, 77.84311



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.73486, 77.82986



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.73651, 77.82982



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.73956, 77.83270



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.80956, 77.85773



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.83351, 77.86931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.83283, 77.87378

